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COMPUTER

CHAPTER 7

Computer System

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EDITION October 2026

LANGUAGE Punjabi & English

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COMPUTER · CHAPTER 7

Computer System

ਕੰਪਿਊਟਰ ਸਿਸਟਮ

IN THIS CHAPTER

- Hardware
- Hardware Vs Software
- Motherboard
- Important Ports At A Glance
- Most Important Facts For Competitive Exams
- One-Line Revision Points

A computer system consists of two main components:

1. Hardware

2. Software

Hardware and software work together to perform tasks. Without hardware, software cannot run, and without software, hardware cannot function effectively.

HARDWARE

Hardware refers to all the physical and tangible components of a computer system that can be seen and touched.
Hardware is the physical part of a computer that we can see, touch, and use.

Examples

- Monitor
- Keyboard
- Mouse
- CPU
- Motherboard
- RAM
- ROM
- Hard Disk
- Printer
- Speakers

HARDWARE VS SOFTWARE

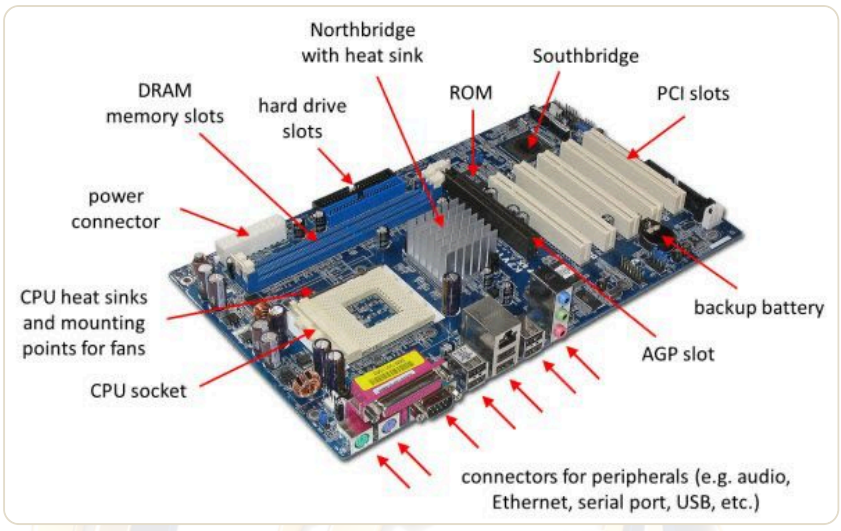
Hardware	Software
Physical components	Programs and instructions
Can be touched	Cannot be touched
Example: Keyboard	Example: MS Word

Hardware	Software
Subject to physical damage	Cannot be physically damaged

MOTHERBOARD

The **Motherboard** is the main circuit board of a computer that connects and holds together almost all internal hardware components.

It acts as the communication center of the computer.



Other Names

- Mainboard
- System Board
- Printed Circuit Board (PCB)

Important Exam Point

Motherboard = Heart/Backbone of Computer System

Functions of Motherboard

1. Connects All Hardware

Connects:

- CPU
- RAM
- ROM
- Hard Disk
- Expansion Cards

2. Supplies Power

Distributes electrical power to all connected devices.

3. Enables Communication

Allows hardware components to communicate with one another.

4. Houses CPU

Provides a socket for installing the processor.

COMPONENTS OF A MOTHERBOARD

1. CPU Socket (CPU Slot)

A special socket provided on the motherboard for installing the CPU.

Exam Point

CPU is installed in the CPU Socket.

2. Memory Slots

Slots used for installing memory modules.

Used For

- RAM
- ROM Modules (in some systems)

Features

- Hold RAM modules (DIMM in desktops, SO-DIMM in laptops)
- Number of slots decides the maximum RAM that can be installed

3. Expansion Slots

Expansion slots allow additional hardware cards to be installed.

Examples

- Graphics Card
- Sound Card
- Network Card
- TV Tuner Card

Exam Point

Expansion Slot = Used to add new hardware capabilities

4. Northbridge

Northbridge is a chipset responsible for communication between:

- CPU
- RAM
- Graphics Controller

Function

Handles high-speed communication inside the motherboard.

Note

Modern processors often integrate Northbridge functions into the CPU itself.

5. USB Port(Universal Serial Bus)

Uses

Connects devices such as:

- Keyboard
- Mouse
- Printer
- Pen Drive
- Scanner

6. PCI Slot(Peripheral Component Interconnect)

Connects:

- Sound Cards
- Modems
- Network Cards
- Video Cards

7. AGP Slot(Accelerated Graphics Port)

Specially designed for: Graphics Cards

Exam Point

AGP = Graphics Card Slot

8. Heat Sink

A metallic component that absorbs and dissipates heat generated by the processor.

Function

Prevents CPU overheating.

Often Used With

- Cooling Fan
- Thermal Paste

9. Power Connector

Supplies electrical power from the SMPS to the motherboard.

10. CMOS Battery(Complementary Metal Oxide Semiconductor)

Stores:

- Date
- Time
- BIOS Settings
- Hardware Configuration even when the computer is switched off.

Exam Point

CMOS Battery maintains BIOS settings.

11. SMPS(Switched Mode Power Supply)

SMPS converts AC power from the wall socket into DC power required by computer components.

Functions

- Controls power supply
- Protects components
- Supplies correct voltage

Exam Point

SMPS = Power Supply Unit of Computer

MICROPHONE

A **Microphone** is an input device that converts sound waves into electrical signals.

Inventor:Emile Berliner in 1877

Uses

- Voice Recording
- Video Conferencing
- Online Classes
- Audio Communication

USB FLASH DRIVE (PEN DRIVE)

A USB Flash Drive is a removable storage device that uses flash memory.

Features

- Portable
- Rewritable
- USB Interface



Uses

- Data Transfer
- Backup
- Portable Storage

Exam Point

Pen Drive uses Flash Memory Technology.

MEMORY CARD

A Memory Card is a portable storage device used to store digital data.

Also Known As

- Flash Memory Card
- SD Card
- MicroSD Card



Uses

- Smartphones
- Cameras
- Camcorders

- Printers
- Gaming Consoles

Types

- SD Card
- MicroSD Card
- MiniSD Card

COMPUTER PORTS

Ports are connection points on a computer where external devices are connected.

Ports act like "gates" through which data enters or leaves the computer.

Types of Computer Ports

1. Hard Disk Ports

A. PATA(Parallel Advanced Technology Attachment)

Features

- Older technology
- Parallel data transfer

B. SATA(Serial Advanced Technology Attachment)

Features

- Faster than PATA
- Widely used today

Exam Point

SATA replaced PATA in modern computers.

C. SCSI(Small Computer System Interface)

Uses

- Servers
- High-performance storage systems
- External devices such as scanners and tape drives

2. Parallel Port

Uses

- Printers
- Scanners

Features

Transfers multiple bits simultaneously.

3. Serial Port

Uses

- Modems
- Projectors

- Older Communication Devices

Features

Transfers data one bit at a time.

Parallel Port vs Serial Port

Parallel Port	Serial Port
Multiple bits at once	One bit at a time
Faster in old systems	Simpler design
Used for printers	Used for modems

4. PS/2 Port

Uses

- Keyboard
- Mouse

Developed By IBM

Exam Point

PS/2 Port = Mouse and Keyboard Connection

5. VGA Port(Video Graphics Array)

Uses

Connects:

- Monitor
- Projector

Signal Type

Analog Video Signal

6. HDMI Port(High Definition Multimedia Interface)

Uses

Transfers:

- Audio
- Video through a single cable.

Devices

- TV
- Monitor
- Projector
- Gaming Console

Exam Point

HDMI carries both Audio and Video.

7. FireWire Port

Developed By Apple

Uses

- High-Speed Data Transfer
- Digital Cameras
- External Storage Devices

Other Name

IEEE 1394

8. Ethernet Port

Used for connecting computers to a network Also Called LAN Port

Uses

- Internet Connection
- Local Area Network (LAN)

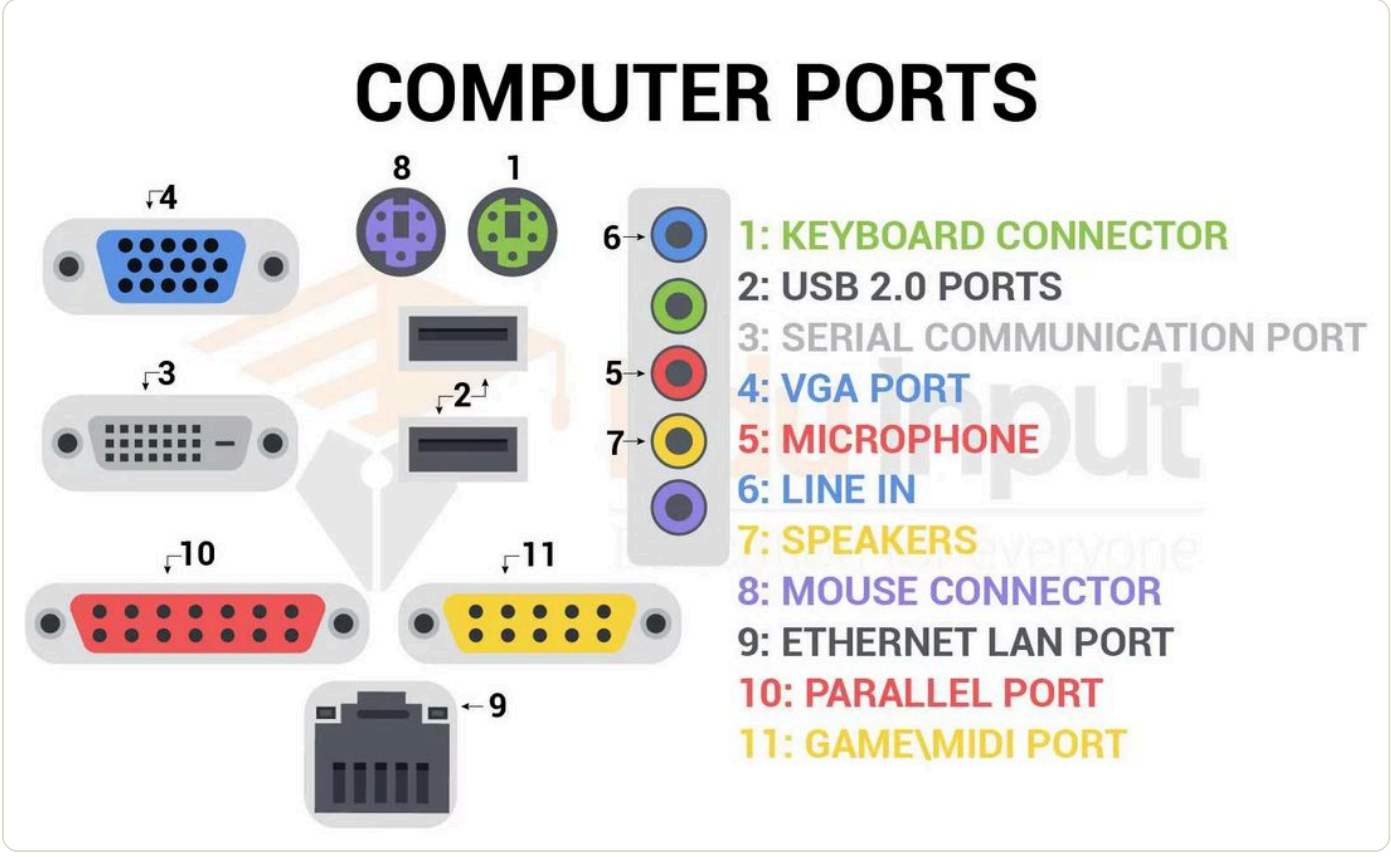
Exam Point

Ethernet Port = LAN Port

9. AGP Port(Accelerated Graphics Port)

Purpose

Dedicated graphics card connection.



IMPORTANT PORTS AT A GLANCE

Port	Full Form	Use
USB	Universal Serial Bus	Mouse, Keyboard, Pen Drive

Port	Full Form	Use
HDMI	High Definition Multimedia Interface	Audio & Video
VGA	Video Graphics Array	Monitor
PS/2	Personal System/2	Keyboard, Mouse
Ethernet	Local Area Network Port	Internet
SATA	Serial Advanced Technology Attachment	Hard Disk
AGP	Accelerated Graphics Port	Graphics Card
FireWire	IEEE 1394	High-Speed Data Transfer

Chipset

A chipset controls communication between:

- CPU
- RAM
- Storage Devices
- Expansion Cards

MOST IMPORTANT FACTS FOR COMPETITIVE EXAMS

- ✔ Hardware refers to physical computer components.
- ✔ Motherboard is called the Heart or Backbone of the Computer.
- ✔ Motherboard is also known as PCB.
- ✔ Intel CPU is installed in CPU Socket.
- ✔ CMOS Battery stores BIOS settings.
- ✔ SMPS controls power supply.
- ✔ USB = Universal Serial Bus.
- ✔ HDMI carries audio and video together.
- ✔ VGA is used for monitors.
- ✔ Ethernet Port is also called LAN Port.
- ✔ SATA replaced PATA.
- ✔ FireWire is also called IEEE 1394.
- ✔ AGP is used for graphics cards.

ONE-LINE REVISION POINTS

- Hardware means physical computer parts.
- Motherboard connects all hardware.
- CPU is installed in CPU socket.
- Expansion slots add new hardware.
- CMOS stores BIOS settings.
- SMPS supplies power.

- USB is the most common port.
- HDMI transfers audio and video.
- Ethernet is used for networking.
- SATA is used for hard drives.
- AGP is used for graphics cards.

QUICK REVISION NOTES (30 SECONDS BEFORE EXAM)

- Hardware = Physical Components
- Motherboard = Heart of Computer
- Main Components:
  - CPU Socket
  - Memory Slots
  - Expansion Slots
  - CMOS Battery
  - SMPS
- Ports:
  - USB → Mouse, Keyboard
  - HDMI → Audio + Video
  - VGA → Monitor
  - PS/2 → Keyboard & Mouse
  - Ethernet → LAN
  - SATA → Hard Disk
  - AGP → Graphics Card
  - CMOS → Stores BIOS Settings
  - SMPS → Power Supply Unit